

Propeller Message Display with Temperature Indicator

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Here we describe a microcontroller-based propeller display that displays any message sent to it via hyper-terminal of a personal computer. Moreover, a tempera-



Fig. 1: Prototype

ture sensing IC (TMP125) is mounted onto the propeller display to display temperature in real time. It gives a 360-degree view and displays several characters in a revolving circular path using just eight LEDs. It is a cost-effective, attractive way of display that reduces the cost and complexity while making the whole system dynamic and energy-efficient. The main highlight of this project is a vertical facing display with true 360-degree viewing angle.

The basic idea of this project comes from the concept of persistence of vision. This phenomenon is related to vision capability of the human eye, where an afterimage is thought to persist for approximately $1/25$ th of a second. So if someone is observing images at a rate of 25 per second, they

appear to form a continuous picture.

In this display, the LED strip moves so fast that one is able to see a matrix of LEDs. The time of a single rotation is divided into several shorter time periods during which each individual LED is kept on/off to display different characters. Temperature sensed by the temperature IC is sent to the controller via SPI bus.

Fig. 2 shows block diagram of the propeller display with temperature indicator. The LED strip is mounted vertically such that when the motor rotates, the strip also rotates in a circular fashion—creating a true 360-degree display. The message to be displayed is sent via RS232 interface using hyperterminal. The message is displayed for 30 seconds and then the display switches to current temperature that is fetched from the temperature sensor. Interrupts are generated by the IR sensor-beam interrupter assembly.

Circuit and working

Fig. 3 shows the circuit of the propeller display. The circuit is built around

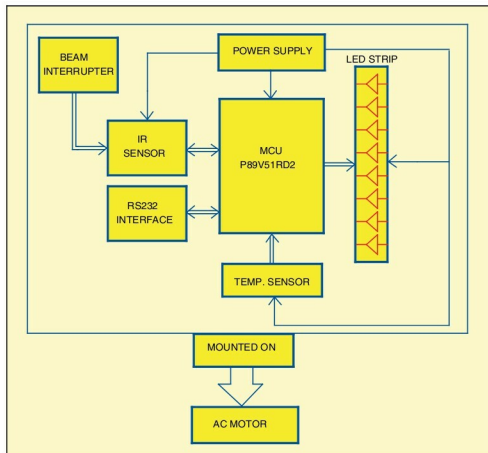


Fig. 2: Block diagram of the propeller message display with temperature indicator



PARTS LIST

Semiconductors:

- | | |
|-----------|-----------------------------|
| IC1 | - P89V51RD2 microcontroller |
| IC2 | - TMP125 temperature sensor |
| IC3 | - MAX232 driver |
| T1 | - BC547 npn transistor |
| IRTX1 | - IR transmitter |
| IRRX1 | - IR receiver |
| LED1-LED8 | - 5mm LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|------------|---------------|
| R1-R8, R12 | - 220-ohm |
| R9, R11 | - 10-kilo-ohm |
| R10 | - 1-kilo-ohm |

Capacitors:

- C1-C5 - 1 μ F, 16V electrolytic
C6, C7 - 33pF ceramic disk
C8 - 10 μ F, 16V electrolytic

Miscellaneous:

- | | |
|----------------|---------------------------------|
| CON1 | - Two-pin connector |
| CON2 | - 9-pin male connector |
| CON3 | - 9-pin female connector |
| COM port | - DB9 connector |
| S1 | - Push-to-on switch |
| X ₁ | - 11.0592MHz crystal oscillator |